Using Machine Learning Algorithms to Predict the State of Financial Inclusion in Africa

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***Abstract*—The financial economy in Africa faces significant challenges that affect development and livelihood. One of these challenges is holding a bank account in Africa, indicating the person's stable economic status. There is a need to solve bank problems in Africa and find solutions to the banking problems. Studies on this topic consider the enormous number of people who do not have a bank account compared to those who have and how this contributes to the decline of Africa's economy. Therefore, in this research, we have implemented effective mechanisms using machine learning techniques to predict who owns a bank account and who is not in African banks. We used different machine learning algorithms, such as SVM, Naive Bays, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Bagging, AdaBoosting, Voting Ensemble, KNN, Stack, and XGBoosting Classifiers. We have experimented with these techniques on a public dataset obtained from African banks (publically available on Zindi) to predict whether a person has a bank account or not. We used the Holdout cross-validation method to split the training dataset randomly to train and validation. The results show that the XGBoost model has a superior accuracy score of 89.23%. This paper provides a comprehensive comparison for all mentioned models, which we used to perform our study.**

***Index Terms*—financial inclusion, bank account, machine learning, classifier, stack.**

I. Introduction

According to the nonpartisan Pew Research Center, Africa is expected to lead the world in population growth by the end of the century [1]. Despite this fact, the World Bank reported, in June 2020, that the COVID-19 pandemic could drive up to 71 million people into extreme hardship that harms fighting poverty [2]. It also estimated the economic impact of the pandemic worldwide, especially in non-developing countries, which leads to a decrease in the income of individuals and families [3]. One of the challenging problems in Africa's economy is the low rate of people who access or use bank accounts. Accessing bank accounts is seen as an indicator of financial inclusion. Also, it is considered an essential contributor to healthy economic growth. It is noticed that having a bank account in Africa depends on gender, type of work, location, and other features. Financial inclusion is essential for an economy to become sustainable [4]. Still, it must be available for everyone to work with initiatives that deliver money. Less than a quarter of adults have access to account banks in Africa, which is considered a tiny percentage [5]. It indicates that many adults use unofficial ways to save their money and borrow money when needed. We believe that knowing if a person owns a bank account is considered an essential step toward solving bank problems for individuals.

In this research, we examine different ML models to achieve precise results to indicate how using a person's features can predict owning an account bank. We performed our analysis on Kenya, Rwanda, Tanzania, and Uganda using a Zindi public dataset [6]. This paper shows a complete comparison of all models which we applied to obtain our prediction. The rest of the article is organized as follows: Section (II) provides the related work. Section (III) presents our methodology to build the best model. Section (IV) shows the results and analysis, and Section (V) concludes the paper.

II. Related Work

Several research papers discussed the bank challenges of Africa. Kanjo et al. [7] presented the services to reach unbanked people to allow them to use bank and financial services without the bank, only using mobile financial services. They stated that they could find unbanked accounts by expanding mobile financial services of some nearby markets and USSD services, a protocol used by GSM cellular mobiles. This study covered the state of banked and unbanked Malawi by a prototype approach using HTML, CSS, JavaScript, MY SQL Database, and PHP programming languages. The goal was to make people use payment services. Demirgüç-Kunt et al. [5] presented a comprehensive study of all African countries regarding financial inclusion and its challenges. They attempted to summarize the financial barriers. For example, in Africa, it is not allowed for women and youths to have bank accounts. The researchers used a novel dataset to determine adults' characteristics that they can have a bank account. Chinaka et al. [8], showed that the lack of financial inclusion is worse in remote areas that depend on agriculture. They

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made a review of challenges of financial inclusion, especially on small-scale farming. The farmers had a shortage in using suitable assets as a guarantee. This work aimed to help these farmers use their right to utilize good assets by applying a new technology to help them, the blockchain. They described how this technology could improve the usefulness of transferring the ownership assets to physical assets. Pearce [9], provided an evaluation state of financial inclusion in the Middle East and North Africa and defined the challenges and constraints to enhance reach to financing. He gave some recommendations and advice to achieve this goal: the government can accept financial inclusion strategy based on improved data. The regulators must provide a legal framework to expand finance reach through the banks and other recommendations.

Several researchers used machine learning to solve such problems. Akinnuwesi et al. [10] applied experimental Machine Learning on Data of financial inclusion in Eswatini. The study's primary goal is to increase the SSB capital base to increase investments, increase job opportunities, and help small business owners have financial and bank services. They made their study in the Eswatini kingdom and used the dataset for 2018 to determine which small-scale businesses had access to money in their business. They divided data into 80% training data and 20% testing data, then they used SVM, which got 69.4% and Logistic Regression with 63.4%. Bazarbash. [11], explained the strong and weak points of machine learning models. It was based on credit assets, one of the solutions that appeared recently in digital technology by having financial technology (FineTech) to increase financial inclusion. Also, they discussed the challenges of credit risk. They used the three most classic machine learning models: Tree-Based models, SVM, and NN. The Tree-Based models, which they used are Decision Trees, Random Forest, and Gradient Boosting Trees.

Our work uses machine learning models to predict if a person possesses a bank account from his features. We have experimented with state-of-the-art machine learning models and show that XGBoost has the best prediction.

III. Methodology

This study is performed on a dataset obtained from Kenya, Rwanda, Tanzania, and Uganda. We aim to shed light on financial economy challenges in Africa using machine learning. We have applied different state-of-the-art machine learning classifiers to find the best model for the dataset.

A. Data

We have used the Zindi public dataset obtained from banks in Kenya, Rwanda, Tanzania, and Uganda (https://zindi.africa/competitions/financial-inclusion-in-africa). The dataset contains a set of features that help determine whether a person has a bank account or not. After exploring the data, we find that the number of people who have a bank account is low. We applied machine learning to identify and decide the appropriate model that determines whether a person in an African country has a bank account or not. The data are clean, and it does not need any processing. Therefore, we fed it to machine learning models.

B. Data Description

This data contains a vast amount of information stored in a train data file, including 13 columns of features. The description of these columns is shown in TABLE I.

TABLE I: Data Description

Variable	Definition
country	Represents the surveyed country which is one of: Kenya, Rwanda, Tanzania, and Uganda
year	Year of the survey
uniqueid	Unique id of each person interviewed
bank_account	If interviewee has bank account" yes or no"
location_type	The type of location such as: Rural, Urban
cellphone_access	If the interviewee has cellphone "yes or no"
household_size	Number of persons who live in the same house
age_of_respondent	Interviewee's age
gender_of_respondent	Interviewee's gender" Female or Male"
relationship_with_head	The interviewee person relationship with the head of the house, it has seven choices:" Parent, Head of Household, Spouse, Child, Other relative, Other non-relatives, Don't know"
marital_status	Interviewee marital status, it has five choices: Married/Living together, Divorced/Seperated, Widowed, Single/Never Married, Don't know
education_level	The highest level of education, it's six choice: No formal education, Primary education, Secondary education, Vocational /Specialised training, Tertiary education, Other/Dont know/RTA
job_type	The type job of interviewee, it has ten choice: Farming and Fishing, Remittance Dependent, Self employed, Formally employed Private, Formally employed Government, Informally employed, Government Dependent, Other Income, No Income, Don't Know Refuse to answer

C. Data Visualization

In this section, we will explore the features and apply a full visualization for analysis

- **bank account:** In Fig.1. "No" means the person has no bank account, and "Yes" means he has a bank account. It is shown, significantly, that the percentage of people who have bank accounts is less than the parentage of people who do not.
- **location type features with bank account:** Fig.2. shows the frequency of people who either own or do not a bank account considering the location type feature. Obviously, the gap between the frequencies is larger in rural places than the urban's.
- **Martial status feature with bank account:** Fig.3. shows the number of persons who have or haven't bank accounts with the location type feature. The location type is divided into Rural and Urban.

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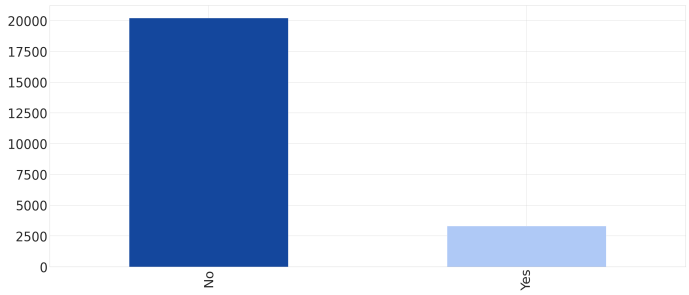


Fig. 1: Number of people having bank accounts

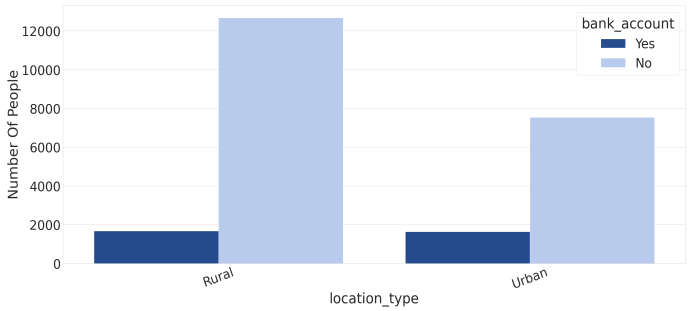


Fig. 2: location type features with bank account

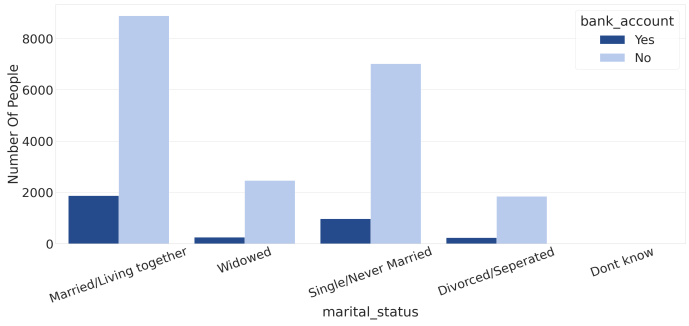


Fig. 3: Martial status feature with bank account

- **Gender of respondent feature with bank account:** Fig.4. shows the number of individuals who have or have not bank accounts concerning the gender of the respondent feature. It can be seen here that the gap between the two numbers for women is larger than for men.

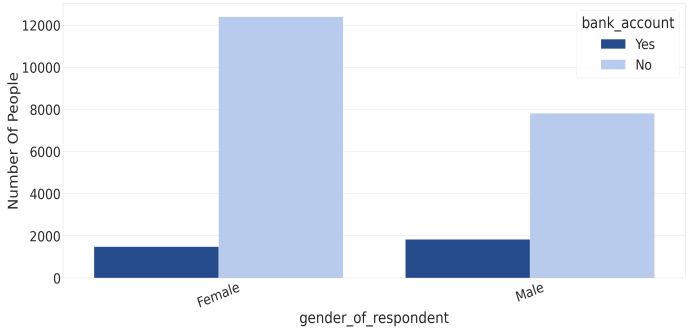


Fig. 4: Gender of respontent feature with bank account

- **Relationship With Head feature with bank account:** Fig.5. shows the number of persons in the study who have or have not bank accounts with relation to his relationship with family's head; the link with the head represents the interviewee person relationship with the house's head. It has seven choices: Parent, Head of Household, Spouse, Child, Other Relative, Other non-Relatives, Don't know.

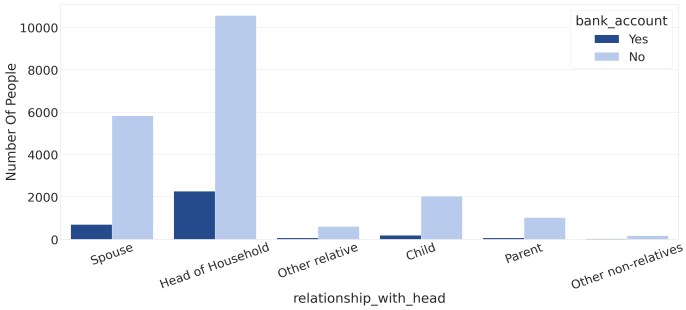


Fig. 5: Relationship With Head feature with bank account

- **Country feature with bank account:** Fig.6. shows the number of persons who have or have not bank accounts with a Country feature. This feature represents the country where the survey occurred. The countries are Kenya, Rwanda, Tanzania, and Uganda.

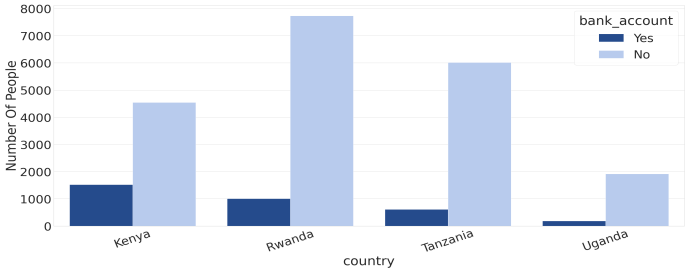


Fig. 6: Country feature with bank account

- **Education feature with bank account:** Fig.7. shows the ratio of individuals who have or have not bank accounts with the education level feature. The education feature represents the highest level of education for that person. It has six choices: No formal education, Primary education, Secondary education, Vocational/Specialised training, Tertiary education, Other/Don't know/RTA.

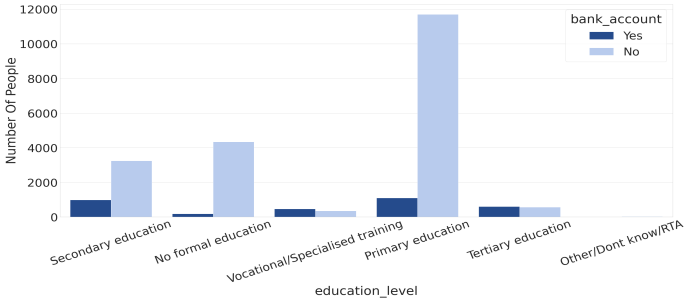


Fig. 7: Education feature with bank account

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- **Job type feature with bank account:** Fig.8. shows the number of persons who have or have not bank accounts with a Job-type feature, representing the interviewee's job type. It has ten choices: Farming and Fishing, Remittance Dependent, Self-employed, Formally employed Private, Formally employed Government, Informally employed, Government Dependent, Other Income, No Income, Don't Know/Refuse to answer.

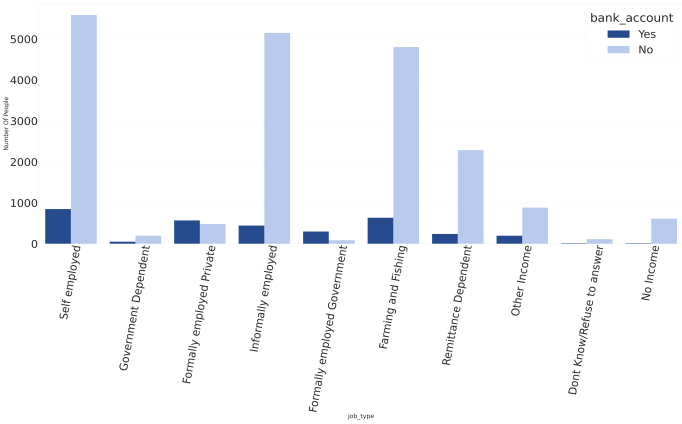


Fig. 8: Job type feature with bank account

- **Cellphone access feature with bank account:** Fig.9 shows the number of interviewed people who have or have not bank accounts with the cellphone feature (the interviewee has a cellphone or not). There are four possibilities; have a bank account and a cellphone, have a bank account and no cellphone, do not have a bank account but have a cellphone, and do not have a bank account or a cellphone. It is found that it is rare to have a bank account and no cellphone.

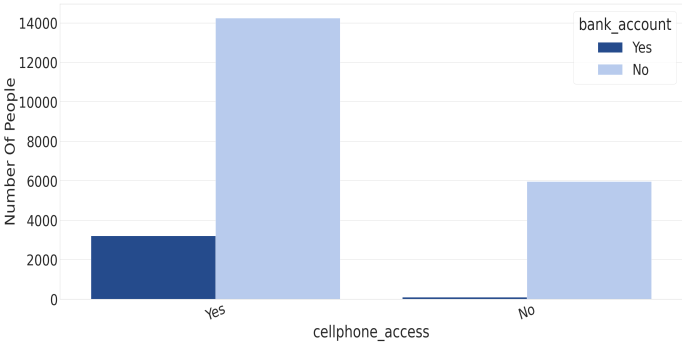


Fig. 9: Cellphone access feature with bank account

- **Data Scatter Plot:**
Fig. 10 shows the scatter plots of interviewee persons who have a bank account and who have not. The yellow dots represent the persons who have a bank account, and the purple dots represent the persons who do not have a bank account. From this figure, we notice that the number of persons who do not have a bank account is enormous compared to the number of persons with a bank account. This is the scatter plot between all features with a label bank account.

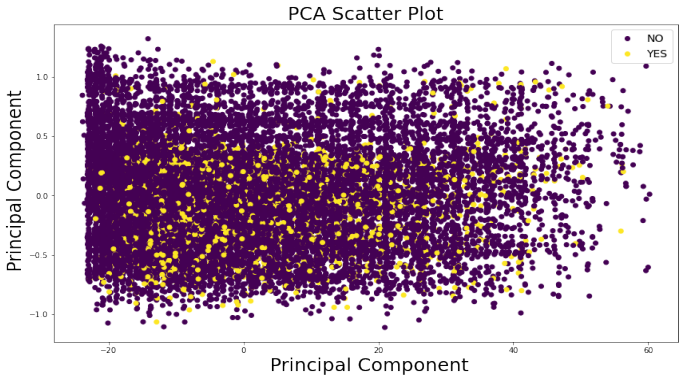


Fig. 10: Data Scatter Plot which yellow dots are persons have bank account and purple dots are persons not have bank account.

D. Rank the features using Random Forest

It is essential to rank the features to explore the highest and lowest correlations between having a bank account and the available features. We used a random forest classifier algorithm with feature_importances attribute to rank these features. TABLE II shows the features' rank using Random Forest. The features are ranked in descending order from highest rank to the lowest rank. It is shown that the Education level reached the highest score. This means that the Educational_level is the most important feature, and it makes a high impact on the prediction more than the others. The Martial_status occupied the lowest score, but also it is an important feature. It is always preferable to use all the data, regardless of the rank.

TABLE II: The rank of features using Random Forest

Feature	**Score**
Education_level	0.34939
Cellphone_access	0.1347
Age_of_respondent	0.11587
Job_type	0.11326
Country	0.08029
Location_type	0.04387
Year	0.03893
Household_size	0.03716
Relationship_with_head	0.03451
Gender_of_respondent	0.02876
Marital_status	0.02325

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E. Highest Performed Model

We have experimented with different machine learning algorithms to predict if a person owns a bank account or not. These algorithm classifiers are SVM, Naïve Bays, Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Bagging, AdaBoosting, Voting Ensemble, KNN, Stack, and XGBoosting Classifiers.

The highest performed model in our experiments is XGBoost Model. XGBoosting is an ensemble machine learning algorithm that is based on Decision Trees and uses gradient Boosting. We used XGBoosting with different parameters to attain the best parameters, as shown in TABLE III.

TABLE III: The Best Parameters of Machine Learning Algorithms

ML Algorithm	Best Parameter
XGBoosting	learning_rate=0.1 n_estimators=200 max_depth=4 min_child_weight=2 subsample=0.7 colsample_bytree=0.6 objective='binary:logistic'
Gradient Boosting	max_depth=3 n_estimators=200 random_state=42
Stack	estimators= Random Forest, XGBoosting, GaussianNB
ADaBoosting	n_estimators=200 random_state=42
Random Forest	bootstrap=True max_depth=8 n_estimators=200 n_jobs=-1
Bagging	base_estimator= Decision Tree
Voting Ensemble	Random Forsest, XGBoosting, Bagging, AdaBoosting
Decision Tree	criterion="gini" max_depth=3
Logistic Regression	C=7.0
KNN	n_neighbors=7
SVM	C=5 KERNAL=RPF

IV. Results and Discussion

For all ML algorithms, we have calculated the Accuracy, Precision, Recall, F1-score. TABLE IV shows the evaluation results of the models. The models are in descending order from highest accuracy to the lowest accuracy. The XGBossting gained the highest accuracy with 89.23%, and NB got the most insufficient accuracy with 84.00%.

TABLE IV: RESULTS OF DIFFERENT MACHINE LEARNING ALGORITHM

ML Model	Accuracy	Precision	Recall	F1- score
XGBoosting	89.23%	0.72	0.39	0.51
Gradient Boosting	89.03%	0.73	0.36	0.48
Stack	88.99%	0.72	0.37	0.49
ADaBoosting	88.69%	0.70	0.35	0.47
Random Forest	88.64%	0.77	0.28	0.42
Bagging	88.61%	0.45	0.56	0.50
Voting Ensemble	88.58%	0.75	0.30	0.42
Decision Tree	88.01%	0.72	0.25	0.37
Logistic Regression	87.56%	0.68	0.24	0.35
KNN	87.14%	0.61	0.26	0.36
SVM	86.19%	0.72	0.041	0.079
NB	84.00%	0.45	0.56	0.50

Fig. 11 shows the precision evaluation metric for the experimented ML models. The precision is the ratio between true positive and all the positives.The equation of Precision is:
Precision = TP/TP+FP.
We found that the highest precision is obtained by applying the Random Forest model with 0.77, and the worst precision is gained from using the Bagging Classifier and Naive Bays models with 0.45.

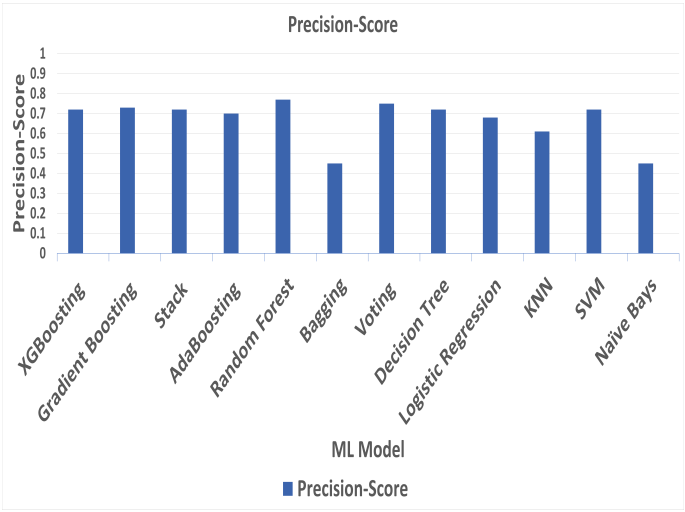


Fig. 11: Precision for ML models

Fig. 12 shows the Recall results of the different experimented machine learning models. The Recall is defined as how accurately the model identifies true positives: the number of true positives divided by the number of true positives plus the number of false negatives. This is the equation of Recall:
Recall = TP / (TP + FN)

We obtained the best Recall by applying Bagging and Naive Bayes models with 0.51 and the worst Recall value by the SVM model with 0.041.

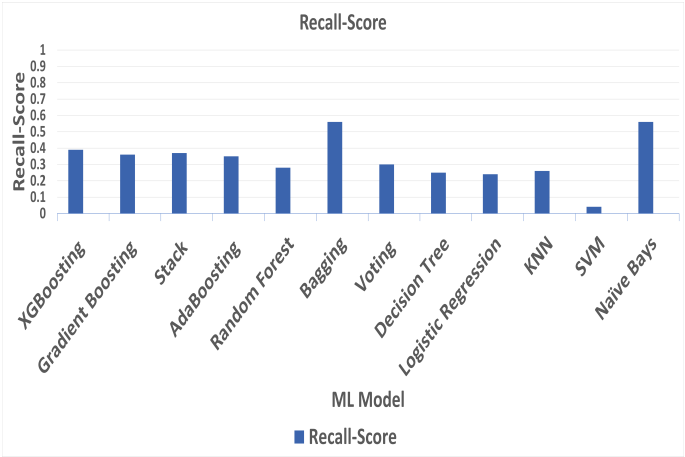


Fig. 12: Recall for ML models

Fig. 13 shows the F1-score of the different machine learning models. F1-score is the harmonic mean of Precision and

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Recall.The equation of F1-score:
F1 Score = 2*(Recall * Precision) / (Recall + Precision)
. The best F1-score model is XGBossting with 0.51, and the worst F1-score by the SVM model with 0.079.

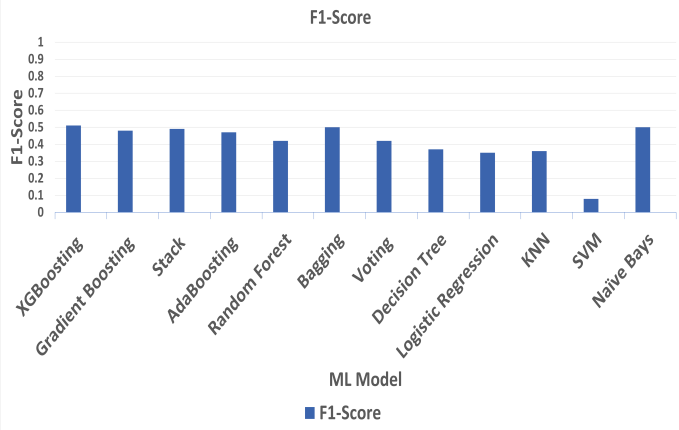


Fig. 13: F1-score for ML models

Fig. 14 shows the Accuracy of different machine learning models, which is the number of correct predicted values from all data. The following is the Accuracy equation:

Accuracy = TP+TN/TP+FP+FN+TN

In the Accuracy term, we got the best result by the XG-Bossting classifier, and we got the worst result by Naive Bayes.

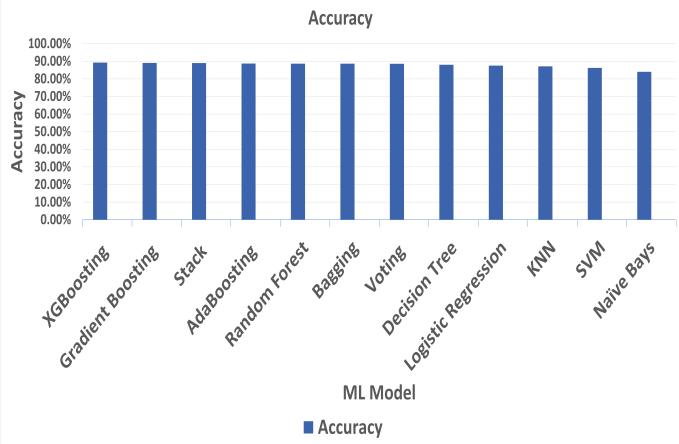


Fig. 14: Accuracy for ML models

We believe that the main challenge in optimizing the accuracy is the imbalanced dataset. Trying under-sampling and over-sampling techniques had decreased the accuracy. Therefore, we did not apply any resampling techniques in our experiments.

The XGBoosting model approaches maximum accuracy, which is equal to ”89.23%”. From the confusion matrix, we can see the False Negative(FN), which means that the number of actual values that are true and were predicted as negative. The False Positive(FP) means that the actual value is negative and was predicted as positive. While the True Positive(TP) shows the correctly predicted positive instances, and the True Negative(TN) shows the correctly predicted negative instances. TP, TN, FN, and FP values are 390, 5908, 611, and 149, respectively. From these values, we can find Recall, Precision, F1-Score, and Accuracy through equations that we discussed before. Precision, Recall, and F1-score results are 0.72, 0.39, and 0.51, respectively. Fig. 15 shows the confusion matrices of the XGBoosting classifier model.

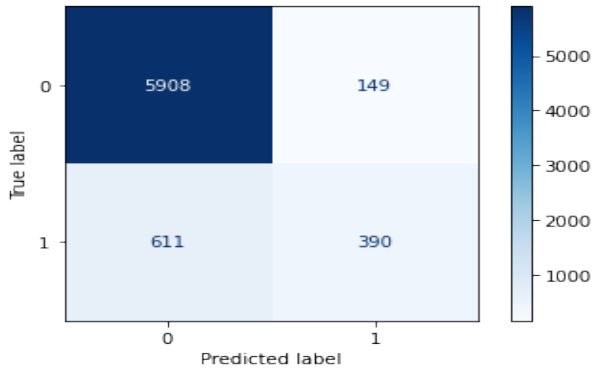


Fig. 15: Confusion Matrix of XGBoosting Model

V. Conclusion

The world is concerned about improving financial inclusion to prevent poverty; however, Africa is suffering in financial inclusion. It faces many problems and challenges in development, leading to a significant deterioration of its economy. This paper sheds light on one of the banking problems: how to know if a person has a bank account. This can help the African banks realize the size of the risks due to this shortage and encourage them to adopt new solutions that accumulate an excellent economy for them and Africa. We have experimented with several state-of-the-art machine learning models to study peoples' features and predict whether they own a bank account or not. We applied different classifier algorithms in machine learning. We found that the best result has been obtained using the XGBoosting classifier, with an accuracy score of 89.23% on a public dataset collected from different African banks.

References

[1] Cilluffo, Anthony, and Neil G. Ruiz. ”World's population is projected to nearly stop growing by the end of the century.” Pew Research Centre 6 (2019): 127.
[2] Mahler, Daniel Gerszon, et al. ”Updated estimates of the impact of COVID-19 on global poverty.” Internet]. World Bank (2020).
[3] United Nations Conference on Trade and Development. ”Impact of the Pandemic on Trade and Development: Transitioning to a New Normal.” (2020).
[4] Bille, Fahima Said, et al. Digital access: the future of financial inclusion in Africa. No. 128850. The World Bank, 2018.
[5] Demirgüç-Kunt, Asli, and Leora Klapper. ”Financial inclusion in Africa: an overview.” (2012).
[6] Zindi, Zindi, [Online]. Available: https://zindi.africa/. [Accessed 20 April 2021].
[7] Kanjo, Chipo, et al. ”ICT solutions for financial inclusion: Reaching out to the unbanked in low resource settings.” 2017 IST-Africa Week Conference (IST-Africa). IEEE, 2017.

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[8] Chinaka, Malvern. Blockchain technology–applications in improving financial inclusion in developing economies: case study for small scale agriculture in Africa. Diss. Massachusetts Institute of Technology, 2016.
[9] Pearce, Douglas. "Financial inclusion in the Middle East and North Africa: Analysis and roadmap recommendations." World Bank policy research working paper 5610 (2011).
[10] Akinnuwesi, Boluwaji A., et al. "Experimental Application of Machine Learning on Financial Inclusion Data for Governance in Eswatini." Conference on e-Business, e-Services and e-Society. Springer, Cham, 2020.
[11] Bazarbash, Majid. "Fintech in financial inclusion: machine learning applications in assessing credit risk." (2019).

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